



ST BENEDICT'S

SUBJECT	CHEMISTRY FINAL EXAMINATION
GRADE	11
EXAMINER	Mr Ntombela
MODERATOR	Mr Hilder
DATE	27 November 2023
MARKS	165
DURATION	2,5 hours

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LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 18 pages, including additional space. A Data Sheet has been provided. Read the questions carefully.
2. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
3. Use the data and formulae sheet whenever necessary.
4. Show your working in all calculations.
5. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
6. Answers must be expressed in decimal format, not left as proper fractions.
7. Where appropriate, express answers to TWO decimal places.
8. It is in your own interest to write legibly and to present your work neatly.
9. A blank page is included at the end of the paper. If you run out of space for a question, use this page. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	Total
Mark									
Moderated Mark									
Question Total	20	30	31	22	19	15	19	9	165

Comment:

QUESTION 1 MULTIPLE CHOICE**20 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	C	D
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Here the option C has been marked as an example.

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D
1.9	A	B	C	D
1.10	A	B	C	D

1.1 Which of the following is the correct formula for ammonium thiosulphate?

- A NH_3SO_3
- B $(\text{NH}_4)_2\text{SO}_4$
- C $\text{NH}_3\text{S}_2\text{O}_3$
- D $(\text{NH}_4)_2\text{S}_2\text{O}_3$

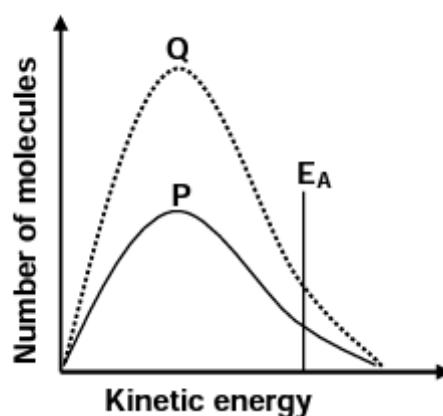
1.2 Which one of the following does NOT contain a dative covalent bond?

- A CO_2
- B SO_2
- C SO_3
- D NH_4^+

1.3 When one mole of gas is at STP the conditions are as follows:

- A It occupies $22,4 \text{ dm}^3$ at STP, i.e. 0°C and at 1 atmosphere.
- B It occupies $22,4 \text{ dm}^3$ at STP, i.e. 25°C and at 101,3 atmospheres.
- C It occupies $24,4 \text{ dm}^3$ at STP, i.e. 0°C and at 101,3 atmospheres.
- D It occupies $24,4 \text{ dm}^3$ at STP, i.e. 25°C and at 1 atmosphere.

- 1.4 Which of the following statements is NOT correct?
- A The strongest type of intermolecular force in ammonia is hydrogen bonding.
 - B The strongest type of intermolecular force in hydrogen chloride is permanent dipole-dipole force.
 - C The strongest intermolecular force in ethanol is permanent dipole-dipole force.
 - D The strongest intermolecular force in water is hydrogen bonding.
- 1.5 Which of the following substances has the greatest number of lone pairs?
- A Nitrogen gas
 - B Oxygen gas
 - C Hydrogen chloride
 - D Hydrogen gas
- 1.6 In giant structures such as diamond and carbon graphite, melting and boiling points are high as a result of:
- A strong covalent bonds between atoms.
 - B strong electrostatic forces between ions.
 - C strong intermolecular forces between molecules.
 - D hydrogen bonding
- 1.7 The Maxwell-Boltzmann distribution curve **P** represents the number of molecules against kinetic energy for a certain reaction.
- Curve **Q** is obtained after a change was made to one reaction condition.



Which ONE of the following changes resulted in curve Q?

- A Addition of a catalyst
- B Increase in temperature
- C Increase in activation energy
- D Increase in the concentration of the reactants

1.8 Which of the following compounds is an alkane?

- A Ethene
- B 2-bromopentane
- C 2-methyl-3,4-dichloropentane
- D 2,3-dimethyl-4-ethyloctane

1.9 Hexan-2-ol undergoes dehydration in the presence of concentrated H_2SO_4 . Which of the following CANNOT be a product of this dehydration?

A	B
C	D

1.10 Which of the following haloalkanes CANNOT undergo a dehydrohalogenation reaction?

- A 2-bromobutane
- B 2,2-dichloropropane
- C trichloromethane
- D 1,2-dibromobutane

QUESTION 2 CHEMICAL BONDING

30 MARKS

2.1 Use only substances from the list below when answering Questions 2.1.1 to 2.1.7.

The phase indicators (state symbols) represent the physical state of each of the substances at room temperature. Substances may be used more than once.)

NaCl (s)	N₂(g)	Fe(s)	NH₃(g)	C: diamond(s)	Ne(g)
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Select one substance from the list that has:

2.1.1 Induced dipole (London) forces between its atoms. (1)

2.1.2 Induced dipole (London) forces between its molecules. (1)

2.1.3 A giant covalent network structure. (1)

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2.1.4 Conducts electricity in its liquid phase. (1)

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2.1.5 Ionic bonds. (1)

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2.1.6 Polar covalent bonds. (1)

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2.1.7 Pure covalent bonds. (1)

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2.2 Carbon and oxygen atoms can bond to form many different compounds.

2.2.1 Define *electronegativity*. (2)

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2.2.2 Draw the Lewis diagram of ethene. (2)

2.2.3 Circle and label a sigma and pi bond in your Lewis diagram in 2.2.2. (2)

2.2.4 Define a dative covalent bond. (2)

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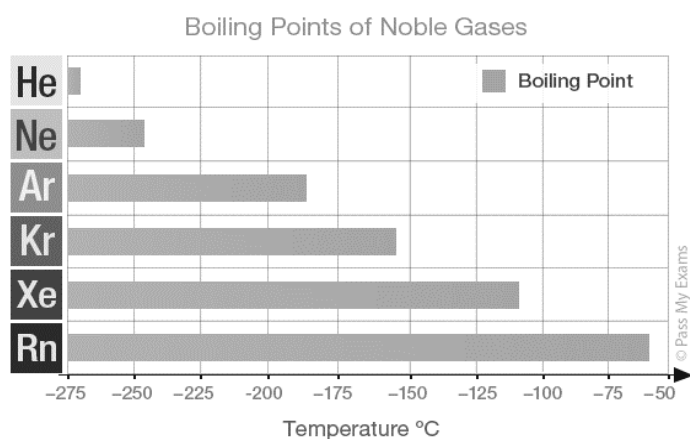
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2.2.5 Draw a Couper diagram of carbon monoxide. (3)

2.2.6 Indicate the polarity of the carbon and oxygen atom in 2.2.5 (2)

- 2.2.7 With reference to electronegativity, intramolecular forces and molecular shape, determine if CO_2 is a polar or non-polar molecule. (5)

- 2.3 Study the graph below.



- 2.4 What phase would Argon be at -200°C ? (1)

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A student concludes that as the size of the atoms increases, so does the boiling point of the noble gas.

- 2.5 With reference to intermolecular forces and the factors influencing their strength, explain the student's conclusion. (4)

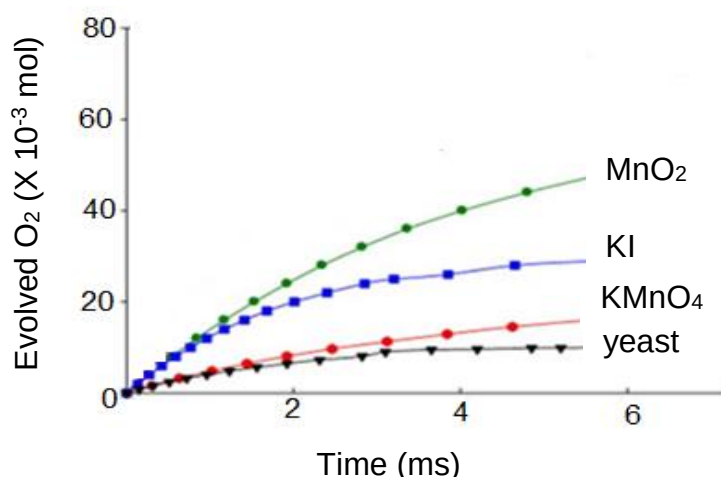
QUESTION 3 REACTION RATES

31 MARKS



Elephant toothpaste is a highly exothermic reaction involving the rapid decomposition of poisonous hydrogen peroxide into water and oxygen gas in the presence of dish soap (to make the foam). The reaction requires a catalyst, which could be yeast, potassium permanganate, manganese (IV) oxide or potassium iodide.

Matthew and Joseph decide to do better, and research which catalyst is the most effective for the reaction. They find the following graph showing various catalysts.



3.1 Define *catalyst*. (2)

3.2 Explain how the data for this experiment would have been collected under laboratory conditions. (3)

3.3 Based on their findings, which of the catalysts would have produced the most foam in the quickest time? (1)

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3.4 Define *rate of reaction*.

(2)

3.5 Using the graph, calculate the average rate of the reaction for the catalyst identified in 3.3 during time 0 – 4 ms. Include appropriate units. (3)

3.6 Matthew says that the reaction is exothermic because energy is only being released. Joseph, wisely, isn't convinced. Explain fully how Matthew's statement needs to be corrected by referring to the breaking and forming of chemical bonds. (3)

3.7 Complete and fully label the energy profile graph below for this reaction. On the same graph, clearly indicate the effect of the addition of yeast in dashed line and the effect of the addition of KI in solid line. (5)



3.8 How would the presence of the catalyst affect the heat of reaction?

Explain your answer fully.

(4)

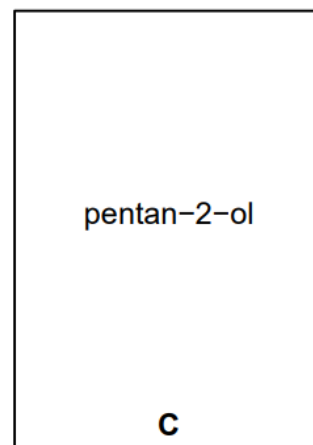
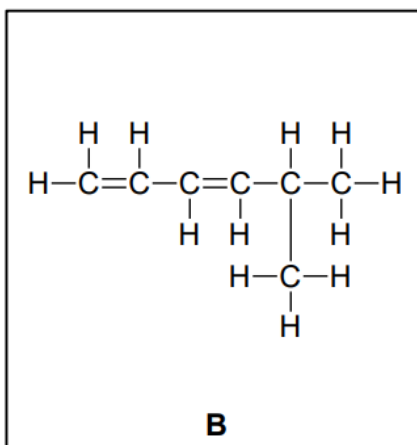
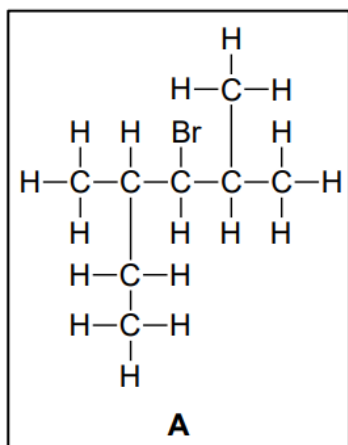
3.9 Fully explain (using bullet points) the effect of an increased concentration of hydrogen peroxide on the reaction by referring to collision theory. (4)

3.10 Mathew and Joseph agree that a temperature change will have an effect on the reaction, draw the corresponding Maxwell-Boltzmann curves showing T1 as the original experiment and T2 as the experiment repeated at a lower temperature. (4)



QUESTION 4 ORGANIC CHEMISTRY

22 MARKS



4.1 Write down the IUPAC name for compound A. (4)

4.2 Draw the condensed-structural formula for compound A. (2)

4.3 NAME the homologous series to which compound B belongs. (1)

4.4 Draw the structural formula for the **unbranched CHAIN** isomer of compound B. (2)

4.5 NAME the functional group present in compound C. (1)

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4.6 State what is meant by the term positional isomers. (2)

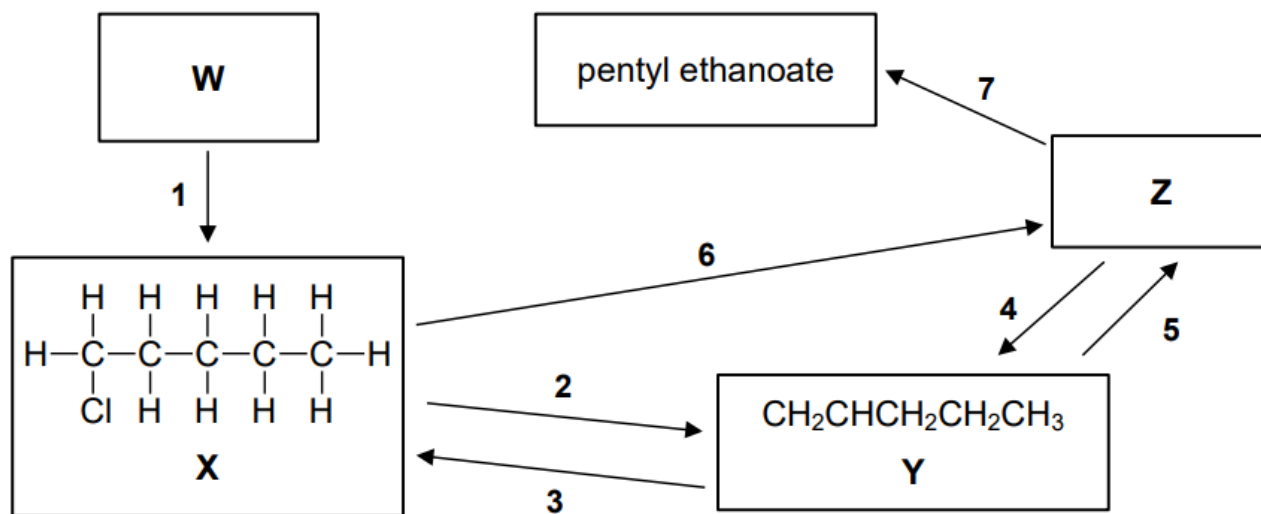
4.7 Draw the structural formula for a POSITIONAL isomer of compound C. (2)

4.8 Compound C has a significantly higher viscosity than compound B. Account for this difference with reference to the predominant intermolecular forces in each of these compounds. (4)

4.9 Using molecular formulae, write a balanced chemical equation for the complete combustion of compound B. (4)

QUESTION 5 ORGANIC CHEMISTRY**19 MARKS**

Consider the flow chart below showing the interconversion of organic molecules through organic reactions, and answer the questions that follow. **W** to **Z** are organic compounds, and 1 to 7 are organic reactions.



The conditions for reactions 1, 5 and 7 are tabulated below.

Reaction	Conditions
1	Cl_2 gas is bubbled through compound W in the presence of UV light.
5	Compound Y is treated with steam and $\text{H}_3\text{PO}_4(\text{aq})$.
7	Compound Z is heated gently with a carboxylic acid, together with a few drops of concentrated $\text{H}_2\text{SO}_4(\text{aq})$.

5.1 Define *hydrocarbon*. (2)

5.2 Write down the IUPAC name of compound **Y**. (2)

5.3 Classify compound **Y** as SATURATED or UNSATURATED. (1)

5.4 Write down the IUPAC name for hydrocarbon **W**. (2)

5.5 Identify the general TYPE of reaction for each of the following:

5.5.1 Reaction 1 (2)

5.5.2 Reaction 2 (2)

5.5.3 Reaction 5

(2)

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5.6 NAME the specific type of addition reaction represented by reaction 3. (1)

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5.7 Consider reactions 5, 6 and 7 and write down the IUPAC NAME of compound Z. (2)

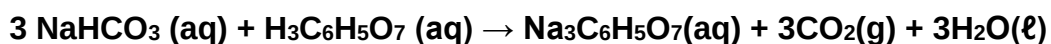
5.8 Draw the structural formula of pentyl ethanoate. (2)

5.9 Write down the **NUMBER** of the reaction that would take place if the relevant organic compound were heated strongly in the presence of concentrated alcoholic potassium hydroxide. (1)

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QUESTION 6**15 MARKS**

The fizz produced when an antacid dissolve in water is caused by the reaction between sodium hydrogen carbonate (NaHCO_3) and citric acid ($\text{H}_3\text{C}_6\text{H}_5\text{O}_7$). The balanced equation for the reaction is:



- 6.1 Write down the FORMULA of the substance that causes the fizz when the antacid dissolves in water. (1)

A certain antacid contains 1,8 g of $\text{H}_3\text{C}_6\text{H}_5\text{O}_7$ and 3,36 g of NaHCO_3 . The antacid is dissolved in 100 cm^3 distilled water in a beaker.

- 6.2 Describe what 1 mole of a substance is. (1)

- 6.3 Calculate the number of moles of NaHCO_3 in the antacid. (3)

- 6.4 Determine, using calculations, which substance is the limiting reagent. (4)

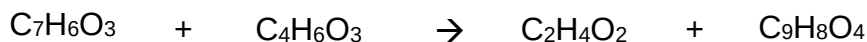
6.5 Calculate the mass of the reactant in excess. (3)

6.6 Calculate the mass decrease of the beaker contents on completion of the reaction. (3)

QUESTION 7**19 MARKS**

Acetylsalicylic Acid (also known as Aspirin) can be made in a microwave oven by reacting salicylic acid with acetic anhydride for 5 minutes. The by-product is acetic acid (also known as vinegar).

Salicylic acid + Acetic anhydride → acetic acid + acetylsalicylic acid (aspirin)



7.1 3,86 g of salicylic acid was reacted with an excess of acetic anhydride.

7.1.1 Define the term *molar mass*. (2)

7.1.2 Calculate the molar mass of a salicylic acid molecule (4)

7.1.3 Calculate the mass of aspirin that should have been produced by 3,86 g of the salicylic acid mass produced. (5)

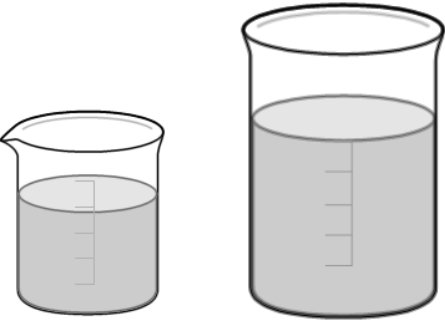
In reality, only 4,65 g of aspirin was produced.

7.1.4 Calculate the percentage yield of aspirin. (3)

- 7.2 A single aspirin tablet contains 300 mg of aspirin. Calculate the number of hydrogen atoms in the 300 mg tablet. (5)
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QUESTION 8**9 MARKS**

Consider the following two aqueous solutions:

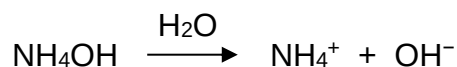
Solution A $c = 0,2 \text{ mol} \cdot \text{dm}^{-3}$ ammonium hydroxide $V = 0,25 \text{ dm}^3$		Solution B $c = 0,3 \text{ mol} \cdot \text{dm}^{-3}$ $(\text{NH}_4)_2\text{CO}_3$ $V = 0,5 \text{ dm}^3$
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8.1 Define the term *solvent*. (1)

8.2 Identify the solvent used in the above solutions. (1)

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The balanced dissociation reaction for ammonium hydroxide is given as:



8.3 Write balanced chemical equation showing the dissociation of ammonium carbonate. (3)

8.4 Solution A is now mixed with Solution B in a new bigger beaker.
Calculate the new concentration of the **ammonium ions** in the solution. (4)

END OF ASSESSMENT

ADDITIONAL SPACE

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE
ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED

[illegible]

[illegible]